10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the jth security's "*beta*" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

a. Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?

ANS:

```
> dat$MSFT_RF <- dat$msft - dat$riskfree
> dat$MKT_RF  <- dat$mkt  - dat$riskfree
>
> # (a) OLS CAPM
> ols_a <- lm(MSFT_RF ~ MKT_RF, data = dat)
> summary(ols_a)

Call:
lm(formula = MSFT_RF ~ MKT_RF, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-0.27424 -0.04744 -0.00820  0.03869  0.35801

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.003250   0.006036   0.538    0.591
MKT_RF      1.201840   0.122152   9.839   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523,    Adjusted R-squared:  0.3486
F-statistic:  96.8 on 1 and 178 DF,  p-value: < 2.2e-16

> confint(ols_a)
                   2.5 %     97.5 %
(Intercept) -0.008661409 0.01516061
MKT_RF       0.960788169 1.44289134
```

As you can see the beta coefficient estimation for Microsoft is 1.201840 and 95% confidence interval is [0.960788169,1.44289135]. Since the beta coefficient estimation for Microsoft is greater than one, which means its volatility is larger than the market portfolio, that is, more risky relative to the market portfolio.

b. It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \ldots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create *RANK* and obtain the first-stage regression results. Is the coefficient of *RANK* very significant? What is the R^2 of the first-stage regression? Can *RANK* be regarded as a strong IV?

ANS:

```
> # (b) RANK and first stage
> dat$RANK <- rank(dat$MKT_RF, ties.method = "first")
>
> fs_b <- lm(MKT_RF ~ RANK, data = dat)
> summary(fs_b)

Call:
lm(formula = MKT_RF ~ RANK, data = dat)

Residuals:
      Min        1Q    Median        3Q       Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.903e-02  2.195e-03   -36.0   <2e-16 ***
RANK         9.067e-04  2.104e-05    43.1   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,    Adjusted R-squared:  0.9121
F-statistic:  1858 on 1 and 178 DF,  p-value: < 2.2e-16

>
> F_b <- summary(fs_b)$fstatistic["value"]
> cat("First-stage F using RANK =", F_b, "\n")
First-stage F using RANK = 1857.587
> cat("First-stage R2 using RANK =", summary(fs_b)$r.squared, "\n")
First-stage R2 using RANK = 0.9125559
```

The first-stage result: MKT_RF(hat) = -0.07903 + 0.0009067 * RANK

The instrument variable RANK potentially satisfies the conditions IV1-IV3.

Since the F-statistic is 1858 and p-value < 2.2e-16, which is much lower than the significance level = 0.05, the coefficient of RANK is very significant. Also, the R-squared is 0.9126.

Thus, we can reject the null hypothesis and infer that instrument variable(RANK) is correlated with MKT_RF, which means that RANK is a good IV.

c. Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?

ANS:

```
> # (c) Hausman test using first-stage residual from (b)
> dat$vhat_rank <- resid(fs_b)
>
> hausman_c <- lm(MSFT_RF ~ MKT_RF + vhat_rank, data = dat)
> summary(hausman_c)

Call:
lm(formula = MSFT_RF ~ MKT_RF + vhat_rank, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27140 -0.04213 -0.00911  0.03423  0.34887 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.003018   0.005984   0.504   0.6146    
MKT_RF       1.278318   0.126749  10.085   <2e-16 ***
vhat_rank   -0.874599   0.428626  -2.040   0.0428 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,	Adjusted R-squared:   0.36 
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16

>
> p_c <- summary(hausman_c)$coefficients["vhat_rank", "Pr(>|t|)"]
> cat("Hausman p-value using RANK =", p_c, "\n")
Hausman p-value using RANK = 0.04278906
```

Vhat is residual of the first stage regression in (b), which is the code resid(fs_b).

As you can see, the t value is -2.040 and Pr(> I t I) is 0.0428. The former is smaller than the critical value and the latter is greater than the significance level = 0.01. Therefore, we cannot reject the null hypothesis, that is, there is no enough evidence to infer that the market return is a endogenous variable(which means it may be a exogenous variable).

d. Use *RANK* as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

ANS:

```
> iv_d <- ivreg(
+     MSFT_RF ~ MKT_RF |
+         RANK,
+     data = dat
+ )
>
> summary(iv_d)

Call:
ivreg(formula = MSFT_RF ~ MKT_RF | RANK, data = dat)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.271625 -0.049675 -0.009693  0.037683  0.355579 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003018   0.006044   0.499    0.618    
MKT_RF      1.278318   0.128011   9.986   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared: 0.3508,	Adjusted R-squared: 0.3472 
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16 

> confint(iv_d)
                   2.5 %     97.5 %
(Intercept) -0.008827134 0.01486322
MKT_RF       1.027421458 1.52921503
>
> cat("OLS beta =", coef(ols_a)["MKT_RF"], "\n")
OLS beta = 1.20184 
> cat("IV beta using RANK =", coef(iv_d)["MKT_RF"], "\n")
IV beta using RANK = 1.278318 
```

From (c) we know that we cannot reject Ho, thus there might be no endogenous problem. Therefore, OLS method in (a) and 2SLS method in (d) are all consistent, that is, the estimated coefficients for market return(MKT_RF) are very close to each other(but IV/2SLS is slightly greater). On the other hand, when it comes to efficiency, OLS is more efficient that 2SLS, as you can see standard error for marker return in (a) is 0.122152, which is smaller than standard error in (d) 0.128011. Consequently, IV estimate agrees with our expectations.

e. Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both $RANK$ and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?

ANS:

```
> # (e) POS and first stage using RANK and POS
> dat$POS <- ifelse(dat$MKT_RF > 0, 1, 0)
>
> fs_e <- lm(MKT_RF ~ RANK + POS, data = dat)
> summary(fs_e)

Call:
lm(formula = MKT_RF ~ RANK + POS, data = dat)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.109182 -0.006732  0.002858  0.008936  0.026652 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.0804216  0.0022622  -35.55   <2e-16 ***
RANK         0.0009819  0.0000400   24.55   <2e-16 ***
POS         -0.0092762  0.0042156   -2.20   0.0291 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,	Adjusted R-squared:  0.9139 
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
> test_e <- linearHypothesis(
+     fs_e,
+     c("RANK = 0", "POS = 0")
+ )
>
> test_e

Linear hypothesis test:
RANK = 0
POS = 0

Model 1: restricted model
Model 2: MKT_RF ~ RANK + POS

  Res.Df     RSS Df Sum of Sq      F    Pr(>F)    
1    179 0.43784                                  
2    177 0.03727  2   0.40057 951.26 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
> cat("First-stage R2 using RANK and POS =", summary(fs_e)$r.squared, "\n")
First-stage R2 using RANK and POS = 0.9148844
```

The first stage regression’s result using RANK and POS as IV is shown above.

The F-statistic is 951.26 and p-value is 2.2e-16. The former is much greater than the critical value and the latter is much less than the significance level 0.05. Accordingly, we can reject the null hypothesis and infer alternative hypothesis is true, that is, RANK and POS are adequately strong IV.

The R-squared of the first-stage regression is 0.9149.

f. Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?

ANS:

```
> # (f) Hausman test using first-stage residual from (e)
> dat$vhat_rank_pos <- resid(fs_e)
>
> hausman_f <- lm(MSFT_RF ~ MKT_RF + vhat_rank_pos, data = dat)
> summary(hausman_f)

Call:
lm(formula = MSFT_RF ~ MKT_RF + vhat_rank_pos, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-0.27132 -0.04261 -0.00812  0.03343  0.34867

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.003004   0.005972   0.503   0.6157
MKT_RF         1.283118   0.126344  10.156   <2e-16 ***
vhat_rank_pos -0.954918   0.433062  -2.205   0.0287 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,    Adjusted R-squared:  0.3625
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16

>
> p_f <- summary(hausman_f)$coefficients["vhat_rank_pos", "Pr(>|t|)"]
> cat("Hausman p-value using RANK and POS =", p_f, "\n")
Hausman p-value using RANK and POS = 0.02874032
```

Since the p-value is 0.02874032, which is smaller than the significance level 0.01. As a result, we cannot reject the null hypothesis Ho: x is exogeneous, that is, there might be no endogenous problem.

g. Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

ANS:

```
> # (g) IV using RANK and POS
> iv_g <- ivreg(
+     MSFT_RF ~ MKT_RF |
+         RANK + POS,
+     data = dat
+ )
>
> summary(iv_g)

Call:
ivreg(formula = MSFT_RF ~ MKT_RF | RANK + POS, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27168 -0.04960 -0.00983  0.03762  0.35543 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003004   0.006044   0.497     0.62    
MKT_RF      1.283118   0.127866  10.035   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared: 0.3507,	Adjusted R-squared: 0.347 
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16 

> confint(iv_g)
                   2.5 %     97.5 %
(Intercept) -0.00884329 0.01485031
MKT_RF       1.03250520 1.53373109
>
> cat("OLS beta =", coef(ols_a)["MKT_RF"], "\n")
OLS beta = 1.20184 
> cat("IV beta using RANK =", coef(iv_d)["MKT_RF"], "\n")
IV beta using RANK = 1.278318 
> cat("IV beta using RANK and POS =", coef(iv_g)["MKT_RF"], "\n")
IV beta using RANK and POS = 1.283118 
```

In (a), the OLS estimated beta coefficient is 1.20184. In (d), the IV estimated beta coefficient using only RANK is 1.278318. In (g), the IV estimated beta coefficient using RANK and POS is 1.283118, which is greater than OLS estimated beta.

The IV estimate agrees with our expectation because OLS will underestimate the estimated coefficient if there is any measurement error.

h. Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

ANS:

```
> # (h) Manual Sargan test
> dat$uhat_iv_g <- resid(iv_g)
>
> sargan_reg <- lm(uhat_iv_g ~ RANK + POS, data = dat)
> summary(sargan_reg)

Call:
lm(formula = uhat_iv_g ~ RANK + POS, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-0.26914 -0.04702 -0.00801  0.03771  0.35674

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0022220  0.0126326  -0.176    0.861
RANK         0.0001370  0.0002234   0.613    0.540
POS         -0.0174499  0.0235409  -0.741    0.460

Residual standard error: 0.08103 on 177 degrees of freedom
Multiple R-squared:  0.003103,  Adjusted R-squared:  -0.008162
F-statistic: 0.2754 on 2 and 177 DF,  p-value: 0.7596

> n <- nrow(dat)
> R2_sargan <- summary(sargan_reg)$r.squared
> Sargan_stat <- n * R2_sargan
>
> df_sargan <- 2 - 1
> p_sargan <- 1 - pchisq(Sargan_stat, df = df_sargan)
>
> cat("Sargan statistic =", Sargan_stat, "\n")
Sargan statistic = 0.5584634
> cat("Sargan df =", df_sargan, "\n")
Sargan df = 1
> cat("Sargan p-value =", p_sargan, "\n")
Sargan p-value = 0.45488
>
> if (p_sargan < 0.05) {
+     cat("Reject surplus IV validity at 5% level.\n")
+ } else {
+     cat("Do not reject surplus IV validity at 5% level.\n")
+ }
Do not reject surplus IV validity at 5% level.
```

As you can see, the p-value for sargan test for validity of the surplus IV is 0.45488, which is much greater than the significance level 0.05. Hence, we cannot reject the null hypothesis Ho: The regression model is not useful in predicting e(i.e. All instrument variables are not correlated with the error term or all instrument variables are exogenous).